
Test-Time Refinement Moves Intelligibility Six Times More Than Identity in Masked-Diffusion TTS

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Abstract

1 A diffusion language model buys sequential computation twice: depth, paid once
2 in parameters, and denoising steps, rented per utterance at serve time. We ask
3 whether the rented kind helps every capability equally. We train an iso-parameter
4 width–depth grid of 15 masked-diffusion codec TTS models (3 seeds, 20–125 M
5 non-embedding parameters) and sweep the refinement budget T at inference only,
6 giving a 75-point (w, d, T) surface per metric from 45 trained models. Fitting
7 $\text{err} = E + Aw^{-\alpha} + Bd^{-\beta} + CT^{-\tau}$ pre-registered, test-time scaling is metric-
8 selective: $\Delta\tau = \tau_{\text{WER}} - \tau_{\text{SIM}} = 0.110$, 95% run-level bootstrap CI [0.092, 0.131],
9 excluding zero. The effect that carries practical weight is one of *magnitude*: over
10 $T=1 \rightarrow 16$ refinement moves 0.952 absolute WER (a $6.1\times$ reduction) against
11 0.161 absolute speaker-similarity error ($1.25\times$), a $5.9\times$ asymmetry. It is not a
12 difference in convergence *rate*: on each metric’s own normalised curve the fraction
13 of total gain realised by $T=8$ is 94.6% for WER and 93.7% for identity, identity
14 marginally later. The selectivity persists across a 276 M four-budget replication,
15 $\Delta\tau = +0.1095$, CI [0.0951, 0.1241]. Depth and steps are *not* interchangeable:
16 the substitution form $B(dT^\kappa)^{-\beta}$ is worse than the separable form by $\Delta\text{AICc} =$
17 $+69.3$.

18 1 Introduction

19 A diffusion language model can buy sequential computation twice. It *owns* depth: d transformer
20 blocks, paid once in parameters and then on every forward pass forever. It *rents* refinement steps:
21 T denoising passes over the same sequence, paid per utterance and adjustable at serve time. An
22 autoregressive model has only the first lever; its serial cost per token is fixed by the architecture the
23 moment training ends. If rented serial compute substitutes for owned serial compute, a deployment
24 can trade parameters for steps after the fact — but only for the capabilities that steps actually improve.

25 Zero-shot text-to-speech is an unusually clean probe of that question, because it separates two
26 capabilities that are measured by standard, independent metrics. Intelligibility is compositional: the
27 model must place the right phonetic content in the right order, and word error rate (WER) reads it
28 out. Speaker identity is representational: the model must carry timbre from a short prompt into new
29 content, and speaker similarity (SIM-o) reads that out. The two need not respond to the same knob.

30 We train one iso-parameter width–depth grid of masked-diffusion codec language models and then
31 sweep T at inference only, which yields a full (w, d, T) response surface per metric at zero additional
32 training cost. Two claims were pre-registered before any run. **H-D2**: refinement steps improve
33 intelligibility more than identity, i.e. test-time scaling is metric-selective. **H-D3**: for intelligibility,
34 steps and depth exchange at a measurable rate κ — how many step-doublings buy one depth-doubling.

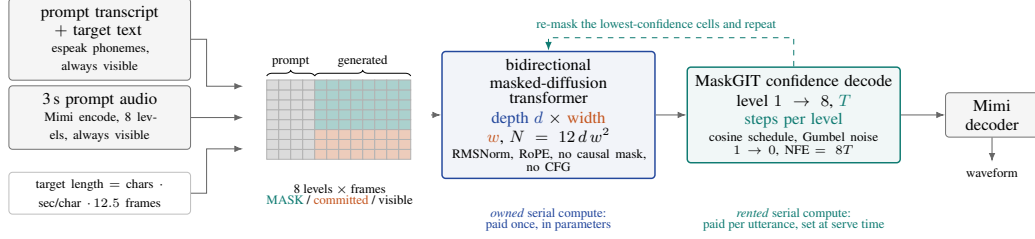


Figure 1: *The two ways a masked-diffusion TTS model buys sequential computation.* Phonemes and a 3-second prompt are always visible; the generated region of the $8 \times \text{frames}$ token grid starts fully masked. The backbone spends *depth* — *owned* serial compute, fixed at training time — and the sampler spends *T refinement steps per codebook level* — *rented* serial compute, chosen at serve time. One training grid over (w, d) therefore yields a whole (w, d, T) surface at inference. This paper asks whether the rented kind helps every capability equally.

$$\text{err}_m(w, d, T) = E_m + A_m w^{-\alpha_m} + B_m d^{-\beta_m} + C_m T^{-\tau_m} \quad \text{err}_m(w, d, T) = E_m + A_m w^{-\alpha_m} + B_m (dT^{\kappa_m})^{-\beta_m} \quad (1)$$

Both claims were registered before the first run, together with the thresholds, the frozen sampler and the analysis plan. The answer to the first is yes and quantitative: $\Delta\tau = 0.110$ with a run-level bootstrap CI of $[0.092, 0.131]$, so the step exponent is reliably larger for intelligibility than for identity. The answer to the second is a clean no: the substitution form is decisively *worse* than the separable one ($\Delta\text{AICc} = +69.3$ for WER, $+71.3$ for SIM-o) and κ runs to its bound, so no finite depth-step exchange rate describes this regime. The two results together say something sharper than either alone: extra inference compute is a targeted instrument, and it cannot be treated as a substitute for architecture.

- ① The first per-metric (w, d, T) scaling surface for speech generation: 75 points per metric from one training grid of 45 models, released with the harness.
- ② $\Delta\tau = 0.110$, CI $[0.092, 0.131]$ — steps are metric-selective — and the *negative* result on depth-step substitution stated as an AICc comparison ($+69.3$) rather than a narrative, with κ reported as unidentified.
- ③ A serving rule measured rather than assumed: c_{layer} is flat in *width* at batch 1 on the evaluation GPU, so serial latency is $8Td_{\text{layer}}$ and the *step budget* is the only latency knob that trades against intelligibility.

2 Related work

Masked diffusion language models are now trained at scale [Nie et al., 2025, Sahoo et al., 2024, Austin et al., 2021], but their scaling is reported against a single likelihood-shaped objective, so a per-capability question cannot arise. In speech, the masked-generative recipes that made codec-token TTS practical — confidence-decoded parallel generation over residual codebook levels [Borsos et al., 2023, Chang et al., 2022] and its multi-stage successors [Wang et al., 2024] — fixed a step budget by engineering judgement and reported a single operating point; none maps error against steps as a law. Autoregressive codec TTS [Wang et al., 2023] has begun to report scaling behaviour, but in total parameters N only [Ye et al., 2025], which cannot separate width from depth and has no step axis at all; classical scaling work is likewise N -indexed [Kaplan et al., 2020, Hoffmann et al., 2022]. Test-time scaling is well studied in reasoning [Snell et al., 2024], where the payoff is measured on one correctness metric; whether extra inference compute helps different capabilities *differently* is, to our knowledge, untested in generative speech. That is the gap this abstract fills: per-metric laws in (w, d, T) for a masked-diffusion TTS model.

3 Method

Backbone and objective. A bidirectional (non-causal) pre-LN transformer over Mimi [Défossez et al., 2024] codec tokens at 12.5 Hz with the first 8 residual codebooks (2048 entries each, plus

MASK and PAD). Width w , depth d , head dimension 64, GELU MLP ratio 4, RMSNorm, RoPE, no weight tying, so the non-embedding parameter count is $N = 12dw^2$. One embedding table per codebook is summed at the input and one linear head per codebook reads out. Training is SoundStorm-style [Borsos et al., 2023] coarse-to-fine masking: per example a level $l \sim \mathcal{U}\{1..8\}$ and a ratio $t \sim \mathcal{U}(0, 1]$; levels below l stay visible, level l is masked i.i.d. with probability t , levels above l are entirely MASK; the loss is cross-entropy on the masked cells of level l only. A phoneme prefix and a 3-second prompt of codec tokens are always visible; there is no speaker encoder, so the backbone itself is the only identity pathway, and no classifier-free guidance, which would add a second inference-time axis and confound the step axis. The target length is fixed before generation from the text length and a corpus character rate (0.0602 s/char).

Frozen sampler. MaskGIT-style confidence decoding [Chang et al., 2022], level by level, with exactly T refinement steps per level, so NFE = $8T$: a cosine schedule sets how many cells remain masked after each step, tokens are drawn categorically at temperature 1.0, and confidences are log-probabilities plus Gumbel noise annealed $1 \rightarrow 0$ within a level; committed cells stay committed. The sampler is identical for every config, seed and T — nothing is tuned per shape or per step budget. Per-item generation noise is keyed so that each item sees the same noise at the first step of every level for every T , making T the only difference between surface points of one run. Before any metric is computed we verify that T reaches the sampler: outputs at $T=1$ and $T=16$ must differ on more than 20% of generated token cells.

Design and statistics. 15 configs span three iso- N budgets $\times$ five aspect ratios, each trained at 3 seeds for 30,000 steps at effective batch 256 with μP [Yang et al., 2021] (base width 256, base LR 0.004); width transfer and depth transfer were verified by five-point LR sweeps before the grid. $T \in \{1, 2, 4, 8, 16\}$ is swept at evaluation only, giving 75 surface points per metric from 45 trained models. Metrics: WER (Whisper-large-v3 [Radford et al., 2023], greedy, the Whisper English normaliser), SIM-o (WavLM-large [Chen et al., 2022] speaker-verification embedding against the *original* prompt waveform, following Anastassiou et al. [2024]), a frozen degenerate-output rule, and UTMOS [Saeki et al., 2022] as a tertiary. Error transforms are $\text{err}_{\text{WER}} = \text{WER}$ and $\text{err}_{\text{SIM}} = 1 - \text{SIM-o}$, computed over all items including degenerates. Both forms of Eq. (1) are fitted by weighted non-linear least squares with 32 multi-starts; $\Delta\tau = \tau_{\text{WER}} - \tau_{\text{SIM}}$, κ , and $\Delta\rho = \alpha_{\text{SIM}}/\beta_{\text{SIM}} - \alpha_{\text{WER}}/\beta_{\text{WER}}$ carry percentile confidence intervals from a bootstrap that resamples at the *run* level — the five T -points of one run share a trained model, so resampling surface points would understate their correlation — with 2000 replicates recomputing every T point.

4 Results

Steps are metric-selective (H-D2, supported). Fitted on all 75 surface points, the step exponent is $\tau_{\text{WER}} = 0.836$ (CI [0.825, 0.851]) against $\tau_{\text{SIM}} = 0.726$ (CI [0.709, 0.744]), so $\Delta\tau = 0.110$ with CI [0.092, 0.131], excluding zero. The raw surface shows it without any model: over the grid, $T=1 \rightarrow T=16$ takes WER from 113.8% to 18.7% (a factor of $6.1\times$ on the error scale) while SIM-o moves only $0.209 \rightarrow 0.370$ ($1.25\times$ on $1 - \text{SIM-o}$), an absolute-error asymmetry of $5.9\times$. The selectivity is not an artefact of scale, training compute, or the metric stack: extending the grid to a fourth budget (276 M non-embedding parameters, ten further runs) gives $\Delta\tau = +0.1095$, CI [0.0951, 0.1241]; retraining three configurations for $3\times$ longer gives $\Delta\tau = +0.1767$, CI [0.0789, 0.2854]; and re-scoring every synthesis under a different ASR or refitting with unbounded log-amplitudes moves it to $+0.1283$ and $+0.1260$ respectively. Extending T to 64 on a 21-run subset shows WER still falling past the grid edge ($0.1516 \rightarrow 0.1294$), so T_{WER}^* on WER’s own scale is 32, confirming the $T=16$ value was a lower bound. The power gate passed in all three budgets before these tests were run.

Depth and steps do not substitute (H-D3, not supported). The pre-registered substitution form collapses: $\Delta\text{AICc}(M_{\text{sub}} - M_{\text{sep}}) = +69.3$ for WER and $+71.3$ for SIM-o, where the hypothesis required $\leq +4$, and the fitted κ sits on its bound for both metrics, i.e. the exchange rate is not identified. We therefore quote no κ . The mechanism is visible in the fits: T enters with a large additive term that decays as $T^{-\tau}$ towards a floor, whereas depth changes the floor itself; a model that forces the two into one factor dT^κ cannot fit both the strong depth ordering at fixed T and the strong T ordering at fixed depth.

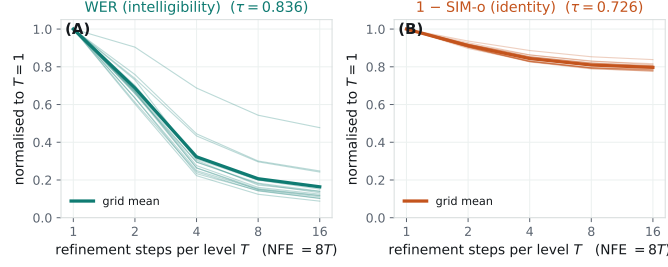


Figure 2: *Test-time scaling is metric-selective.* Error versus **refinement steps** for all 15 configs, each normalised to its own $T=1$ value; thick lines are grid means. (A) WER error falls 0.9518 in absolute terms over $T=1 \rightarrow 16$ and is still falling at $T=16$. (B) SIM-o error falls 0.1606 over the same range, $5.9\times$ less. Same models, same frozen sampler, same items — only T differs.

Table 1: *Fitted exponents* on the 75-point surface (M_{sep} for α, β, τ ; M_{sub} for κ) with 95% run-level bootstrap CIs for the step exponents. $\Delta\tau = 0.110$, CI [0.092, 0.131], excludes zero: the pre-registered primary result. κ is at its bound in both rows and $\Delta\text{AIC}_{\text{csub-sep}} \gg +4$, so no exchange rate is quotable. $\Delta\text{AIC}_{\text{cfull-N}} \leq -4$ means iso- N shape matters.

Metric	α (width)	β (depth)	τ (steps)	κ	$\Delta\text{AIC}_{\text{csub-sep}}$	$\Delta\text{AIC}_{\text{cfull-N}}$
WER	1.574	2.040	0.836 [0.825, 0.851]	3.000 [†]	+69.3	−15.5
SIM-o	0.919	0.553	0.726 [0.709, 0.744]	3.000 [†]	+71.3	−5.5

[†]at the fitted bound, i.e. not identified; $\Delta\tau = 0.110$, CI [0.092, 0.131].

Shape matters, but the anisotropy ratio is not identified (H-D1, not supported). At $T=16$ the two-axis model beats the N -only model for both metrics ($\Delta\text{AICc} = -15.5$ for WER, -5.5 for SIM-o, against a -4 threshold), so error is not a function of parameter count alone — iso- N shape choice matters. The direction of the anisotropy, however, we cannot claim: $\Delta\rho = 0.174$ with CI $[-0.711, 0.388]$, spanning zero, and the width amplitude A saturates the pre-registered bound $A \leq 10$ in every fit, so only $A w^{-\alpha}$ is identified and α — hence $\rho = \alpha/\beta$ — is not. The **depth** exponent, by contrast, *is* identified (B interior in both fits) and separates the capabilities sharply: $\beta_{\text{WER}} = 2.040$ against $\beta_{\text{SIM}} = 0.553$. What the data do show unambiguously is a strong **depth** ordering at fixed parameter count: moving from the shallowest shape to the best-performing depth is worth 40.2, 16.3 and 7.3 WER points in the 20, 50 and 125 M budgets, while SIM-o varies by under 0.065 across all 15 shapes. The depth optimum is *interior* only in the smallest budget ($d=18$ of 4–24); in the two larger budgets the argmin sits at the deepest shape tested ($d=30$ of 6–30 and $d=36$ of 8–36), so those are censored lower bounds and we claim no $d^*(N)$ trend.

Extrapolation and reliability. Fitting the two smaller budgets and predicting the largest gives MAPE 2.4% for SIM-o (versus 2.2% for the N -only model), inside the 15% the hypothesis asked for; WER does not extrapolate (20.2% versus 20.4%). Degenerate outputs — empty, truncated or looping transcripts under the frozen rule — fall monotonically with the step budget, from 40.4% at $T=1$ to 0.1% at $T=16$: most of what $T=1$ costs is not subtle quality but outright failures.

Serving corollary. Measured on the evaluation GPU at batch 1, c_{layer} is flat in width across the whole grid (0.493–0.514 ms per layer for $w \in [256, 1152]$): at these sizes the device is launch-bound, so serial latency is $8Td c_{\text{layer}}$ and width is free, so a latency budget constrains only the product Td . These data do not resolve how to split it: the pre-registered rule that depth should be saturated before steps are raised failed (satisfied at 5.9% of 40 log-spaced budgets by the measured method and 8.8% by the fitted one, against an 80% bar), and T is censored at 16 by the tested grid, so no directional allocation rule is claimed in either direction.

Where the steps are spent matters as much as how many (exploratory). At matched total NFE = 32 on the 125 M budget, spending every refinement pass on the coarsest codebook level rather than the finest changes WER from 1.1480 to 0.2191 (paired item-level bootstrap difference -0.9289 , CI $[-0.9595, -0.8999]$) and the degenerate rate from 39.1% to 0.1%; the coarse schedule is

not free, costing 0.3057 against 0.3477 SIM-o and 2.17 against 2.51 UTMOS relative to a uniform split (Appendix C).

5 Limitations and outlook

The ceiling is 133 M non-embedding parameters and 30,000 steps; every model is under-trained by frontier standards, and the T curves may flatten differently at scale — $T_{\text{WER}}^* = 16$ on WER’s own scale is a lower bound, since WER had not saturated at the largest budget we measured. Part of the identity gap is not ours to close: round-tripping the ground-truth target through the codec and scoring it against the prompt gives 0.555 against 0.678 for the original audio, so 62% of the distance between real speech and our best measured system is lost in the codec itself. One language, one codec, one corpus, and no classifier-free guidance — guidance is a second inference-time axis that would confound the step axis, so excluding it is deliberate, and it is the obvious follow-up. The $\Delta\tau$ interval resamples 3 runs per config, so it captures seed-to-seed variation only, not the uncertainty of the single shared harness and corpus. Target length is set from a corpus-median seconds-per-character rather than per-item speaking rate, so an utterance whose speaker is unusually fast or slow is synthesised at the wrong duration; this penalises both metrics and is not corrected for. $\Delta\rho$ and κ are reported as unidentified rather than estimated. Cuts taken: none. This is pre-registered outcome class S1, and selectivity is the only positive claim the data license.

Reproducibility. Protocol, hypotheses, thresholds, frozen sampler and analysis plan were registered before the first run; every gate is logged with its measured values. The released artefact holds the harness, all 45 run records, the 75-point surface, and the fitting and bootstrap code. Total cost 143 GPU-hours.

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225 A The shape grid, as run

226 Every config in the active grid with its measured $T=16$ means (seeds 0, 1, 2, 400 eval_zs items each).
227 Iso- N within a budget: the three budgets sit at 20, 50 and 125 M non-embedding parameters and the
228 five shapes per budget span the aspect-ratio range at fixed $N = 12dw^2$.

config	budget	w	d	heads	N (M)	val loss	WER (%)	SIM-o	Degen (%)
A1	A	640	4	10	19.7	4.638	53.9	0.327	0.67
A2	A	448	8	7	19.3	4.582	28.4	0.348	0.08
A3	A	384	12	6	21.2	4.566	15.9	0.360	0.17
A4	A	320	18	5	22.1	4.574	13.6	0.359	0.00
A5	A	256	24	4	18.9	4.611	15.3	0.351	0.17
B1	B	832	6	13	49.8	4.551	27.8	0.361	0.08
B2	B	640	10	10	49.2	4.505	16.0	0.381	0.00
B3	B	512	16	8	50.3	4.497	11.9	0.388	0.08
B4	B	448	22	7	53.0	4.511	12.0	0.382	0.08
B5	B	384	30	6	53.1	4.520	11.5	0.375	0.08
C1	C	1152	8	18	127.4	4.504	17.1	0.373	0.17
C2	C	960	12	15	132.7	4.504	18.9	0.373	0.17
C3	C	768	18	12	127.4	4.488	14.3	0.392	0.00
C4	C	640	26	10	127.8	4.489	13.4	0.390	0.08
C5	C	512	36	8	113.2	4.487	9.9	0.388	0.00

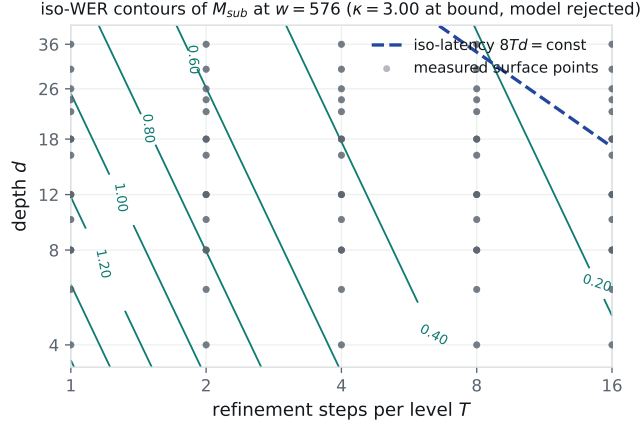


Figure 3: *Why the substitution form is rejected.* Iso-WER contours of M_{sub} in $(\log T, \log d)$ at the median width, with the iso-latency line $8Td = \text{const}$. The fitted κ sits on its bound and $\Delta\text{AICc}(M_{\text{sub}} - M_{\text{sep}}) = +69.3$, so these contours are shown as a rejected model, not as a design rule.

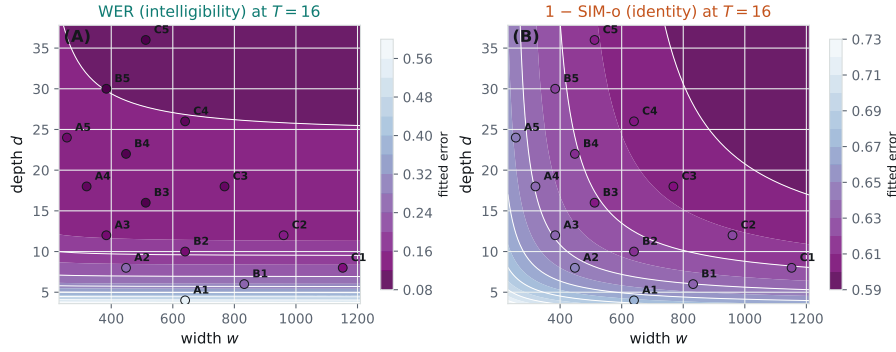


Figure 4: *Fitted anisotropy surface at $T=16$ for both metrics, with the measured config means overlaid.* The width amplitude A saturates its pre-registered upper bound for both metrics, which is why $\rho = \alpha/\beta$ is reported as unidentified in §4 even though the two-axis model beats the N -only model ($\Delta\text{AICc} = -15.5$ and -5.5).

230 B Additional figures

231 C Refinement-budget allocation across codebook levels

232 Three frozen per-level step schedules at matched total NFE = 32 on the 125 M budget (15 runs, 400
 233 items). Paired item-level bootstrap, 2000 replicates, pairing on the eval item so item difficulty cancels.
 234 Coarse spends all passes on codebook level 0, fine on level 7, uniform splits evenly.

schedule	WER	SIM-o	UTMOS	degenerate
coarse [25, 1, ..., 1]	0.2191	0.3057	2.17	0.1%
uniform [4, ..., 4]	0.3144	0.3477	2.51	0.5%
fine [1, ..., 1, 25]	1.1480	0.2246	1.56	39.1%

236 Allocation is metric-selective in the same way the step *count* is: coarse wins WER but loses identity
 237 and naturalness against uniform, both with intervals excluding zero. It is a trade, not a free win.

238 **A significance-based saturation contrast.** Defining saturation as the smallest T whose paired-
 239 bootstrap interval for $\text{err}(T) - \text{err}(64)$ covers zero gives $T_{\text{WER}}^* = 32$ against $T_{\text{SIM}}^* = 16$ on the
 240 extended grid. We report this only here and do not build on it: roughly half the contrast is a precision
 241 artefact, since identity’s detection floor is $\approx 1.8\times$ looser relative to its own dynamic range than

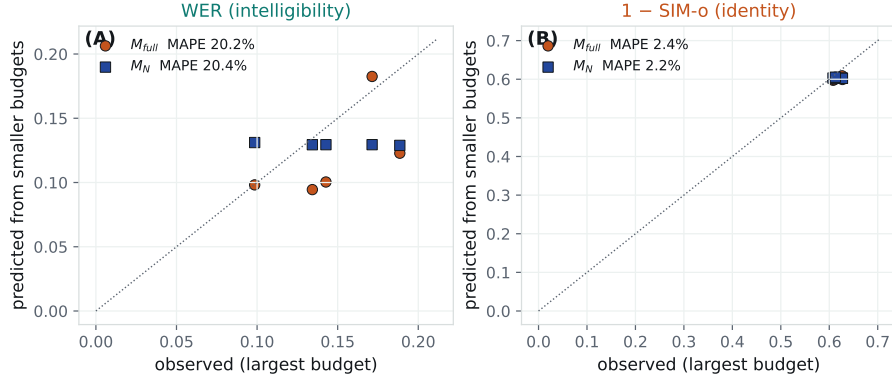


Figure 5: *H-D4 extrapolation*. The two smaller budgets are fitted and the largest budget predicted, under M_{full} and the N -only model. SIM-o extrapolates (MAPE 2.4 % vs 2.2 %); WER does not (20.2 % vs 20.4 %).

intelligibility's. The raw-scale T^* — 95% of the $T=16$ value — is not affine-invariant and must never be compared across metrics: the same 5% band is 0.67% of WER's range but 11.96% of SIM-o's. On the invariant fraction-of-total-gain definition, $T^* = 16$ for both metrics.

D Fit diagnostics and exploratory analyses

Parameters at bounds. The width amplitude A saturates the pre-registered *upper* bound $A \leq 10$ in every fit (Part A: 10.000 for both metrics; Part B: 9.999 for WER, 10.000 for SIM-o), so only the product $A w^{-\alpha}$ is identified and α is not; the bound, not the data, is binding, which is a limitation of the pre-registered parameterisation worth fixing in any follow-up. κ reaches its upper bound in every one of the 2000 bootstrap replicates of M_{sub} . The irreducible term E reaches 0 for WER under M_{sep} . The depth amplitude B is interior throughout, so β is identified.

Exploratory: the depth exponent separates the capabilities. $\beta_{WER} = 2.040$ against $\beta_{SIM} = 0.553$, a factor of 3.7 — the asymmetry H-D1 sought, in the parameter the data pin down. Part-B ρ values point the pre-registered way ($\Delta\rho = +0.890$) but inherit the A -cap problem; the pre-registered test is Part A, so this is exploratory only.

Weighting. Points are weighted $1/SE^2$ with SE taken over seeds, floored at a quarter of the pooled seed SD so that a chance agreement between two seeds cannot dominate the fit; the number of floored points is recorded in the released `fits.json`.

Exploratory (not part of any declared claim). An unweighted refit of both parts is released alongside the weighted fit, as is a fit that additionally includes the finished runs of a stretch third seed. Both reproduce the sign and significance of $\Delta\tau$.

Use of Generative AI Disclosure

Generative AI was used as a coding assistant for the experimental harness and for language refinement of author-written text. The research design, the pre-registered hypotheses and thresholds, the analysis plan, and all scientific claims are the authors' own; the authors verified the code and the numbers and take full responsibility for this paper.